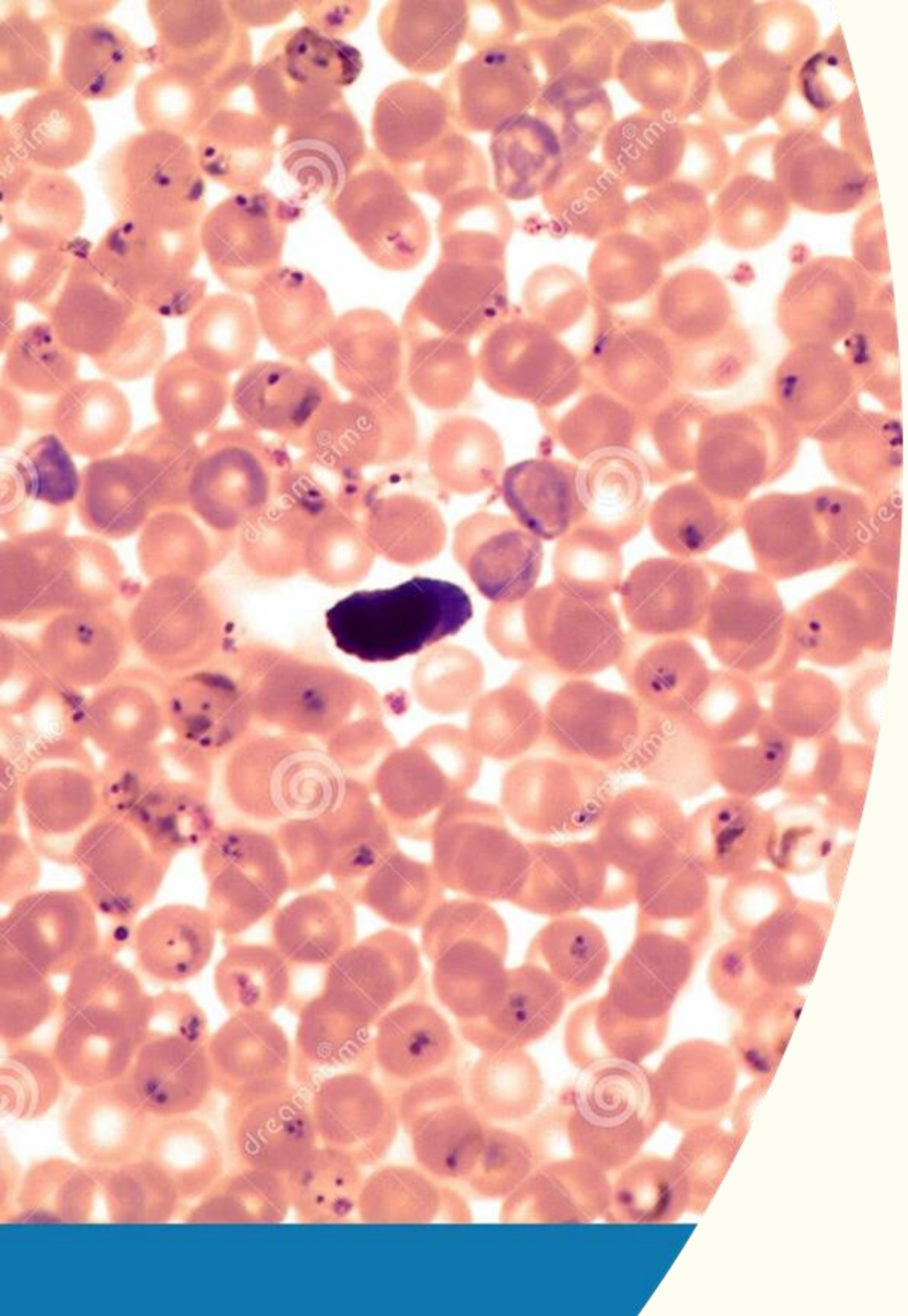

AI-Driven Malaria Detection Using Convolutional Neural Networks

Using MobileNetV2 to classify parasitized vs. uninfected blood smear images

DEEP LEARNING · MEDICAL IMAGING · TRANSFER LEARNING

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Introduction & Motivation

The Global Burden

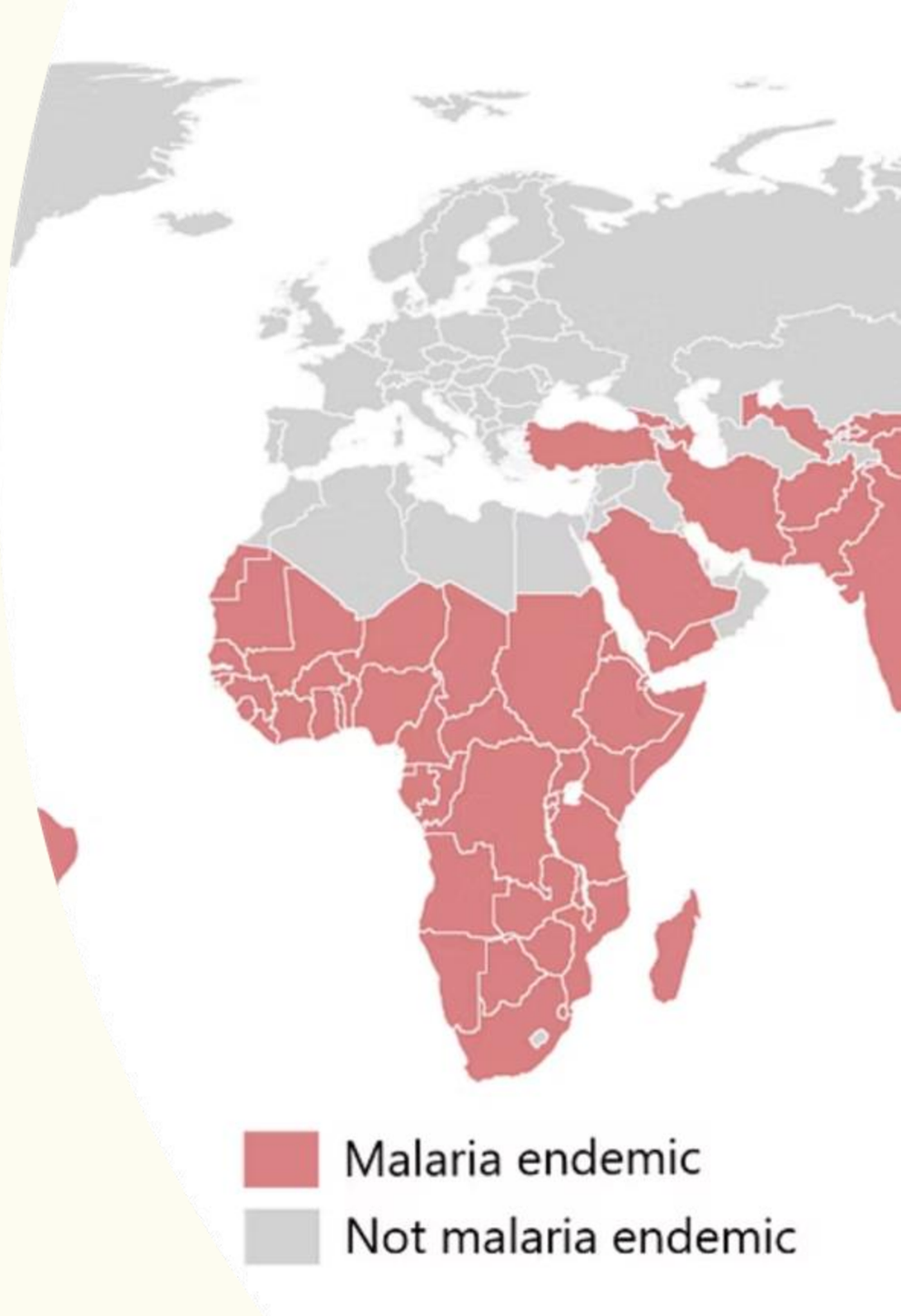
Malaria remains one of the world's most prevalent infectious diseases, claiming hundreds of thousands of lives annually disproportionately in sub-Saharan Africa and Southeast Asia.

The Diagnostic Bottleneck

The current gold standard manual microscopic examination of stained blood smears is highly accurate but slow, labor-intensive, and dependent on trained technicians. In low-resource settings, this bottleneck costs lives.

Our Goal

Build a deep learning model that automatically classifies individual red blood cell images as **Parasitized** (Plasmodium-infected) or **Uninfected**, enabling faster, scalable, and more accessible diagnosis.



Recent Advances in AI for Malaria Diagnosis

1

AI FOR DIAGNOSIS

Performance & Efficiency

Rajaraman et al. (2018): Ensemble CNNs achieved >97% accuracy on NIH dataset; MobileNet competitive and resource-efficient.

Masud et al. (2020): Lightweight custom CNN yielded 97.13% accuracy with <2M parameters, ideal for mobile deployment.

2

TRANSFER LEARNING

Leveraging Pre-trained Models

Talo (2023): Crucial for medical imaging with limited data, fine-tuning general features from ImageNet for domain-specific patterns.

MobileNetV2 (Sandler et al., 2018): Features inverted residuals and linear bottlenecks, offering state-of-the-art performance with minimal computational footprint.

3

CLINICAL INTEGRATION

Challenges in Resource-Limited Settings

Odekunle et al. (2020): Emphasized the need for technical accuracy, robustness, and simplicity, considering infrastructure, user training, and integration with existing health systems.

4

GAPS & OPPORTUNITIES

Addressing Key Limitations

Many studies lack focus on **model interpretability**, which is essential for clinical acceptance.

Limited research on optimizing models for the specific **hardware constraints** of rural health facilities.

Problem Statement & Dataset

Binary Classification Task

Given a microscopic image of a single red blood cell, predict whether it contains the **Plasmodium parasite**. Each image maps to exactly one of two labels: Parasitized or Uninfected.

NIH Public Dataset

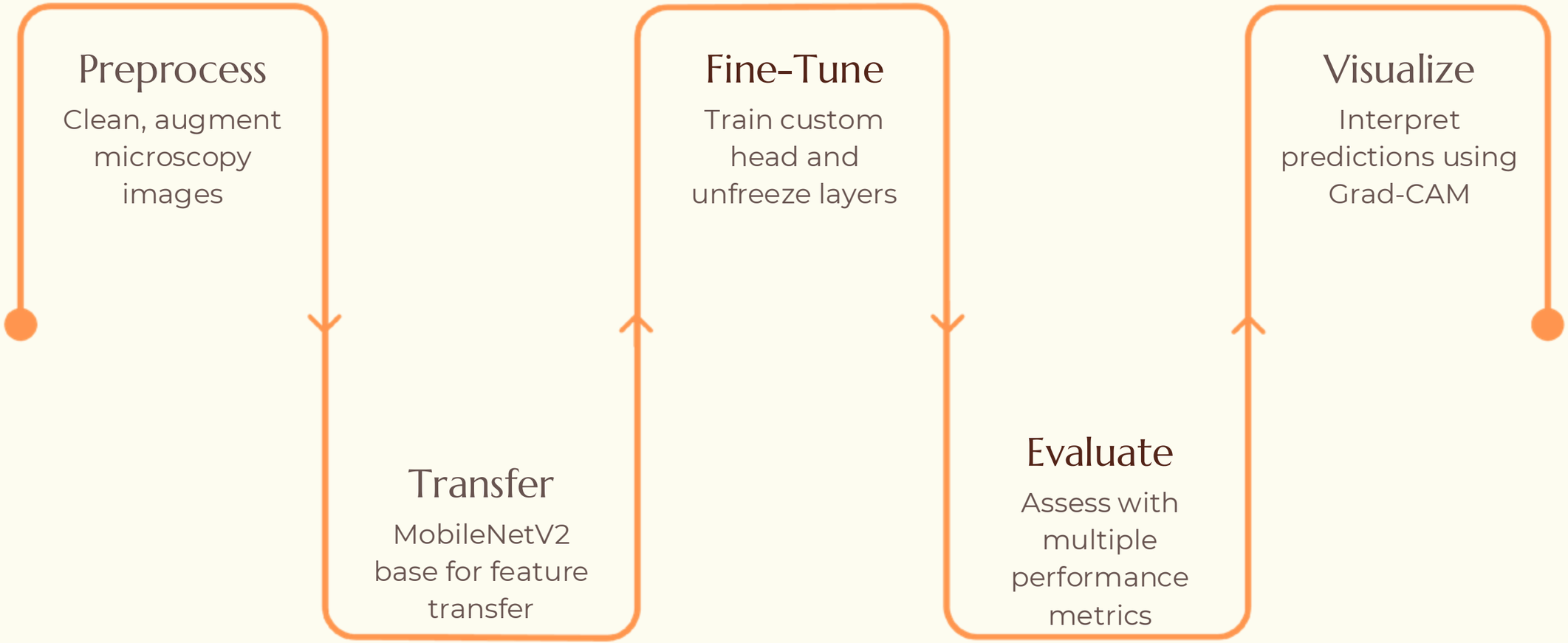
Sourced from the publicly available NIH malaria cell image repository. Images were unzipped from Google Drive and subsampled to **10,000 images** (5,000 per class) to keep training computationally tractable.

Balanced & Split

The dataset is perfectly balanced — critical for unbiased accuracy metrics. It is partitioned into **80% training** (8,000), **10% validation** (1,000), and **10% test** (1,000) sets for rigorous evaluation.

End-to-End Pipeline Overview

The project implements a complete workflow from raw microscopy images to a trained, interpretable classifier — structured as five sequential stages.



Transfer learning allows the model to leverage rich visual features pre-learned from ImageNet's millions of images, dramatically reducing the data and compute requirements for a medical imaging task. MobileNetV2's lightweight architecture also makes eventual deployment on edge devices — such as field tablets or smartphones — entirely feasible.

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 128, 128, 3)	0	-
Conv1 (Conv2D)	(None, 64, 64, 32)	864	input_layer[0][0]
bn_Conv1 (BatchNormalizatio…	(None, 64, 64, 32)	128	Conv1[0][0]
Conv1_relu (ReLU)	(None, 64, 64, 32)	0	bn_Conv1[0][0]
expanded_conv_dept… (DepthwiseConv2D)	(None, 64, 64, 32)	288	Conv1_relu[0][0]
expanded_conv_dept… (BatchNormalizatio…	(None, 64, 64, 32)	128	expanded_conv_de…
expanded_conv_dept… (ReLU)	(None, 64, 64, 32)	0	expanded_conv_de…
expanded_conv_proj… (Conv2D)	(None, 64, 64, 16)	512	expanded_conv_de…
expanded_conv_proj… (BatchNormalizatio…	(None, 64, 64, 16)	64	expanded_conv_pr…
block_1_expand (Conv2D)	(None, 64, 64, 96)	1,536	expanded_conv_pr…

Data Preprocessing & Augmentation

Training Generator

The `ImageDataGenerator` rescales pixel values to [0, 1] and applies a suite of random transformations to each training batch:

- Random rotations and width/height shifts
- Zoom and horizontal/vertical flips
- Shear transformations

Validation & Test Generators

Only pixel rescaling to [0, 1] — **no augmentation**. This ensures evaluation is performed on clean, unmodified images that reflect real deployment conditions.

Why Augment?

Augmentation artificially expands the effective training set, reduces overfitting, and teaches the model to be invariant to the natural variations found in real-world microscopy slides (orientation, zoom level, staining intensity).

Model Architecture: MobileNetV2 + Custom Head

1

MobileNetV2 Base (Frozen)

Loaded **without the top classification layer** (`include_top=False`). All base weights are frozen (`base_model.trainable = False`), preserving the rich ImageNet feature representations.

2

GlobalAveragePooling2D

Reduces each feature map to a single scalar, dramatically cutting the number of parameters compared to a Flatten layer while retaining spatial context.

3

Dense(128, relu) + Dropout(0.5)

A fully connected layer introduces task-specific learning capacity. A 50% dropout rate prevents overfitting on the limited medical image data.

4

Dense(1, sigmoid) Output

Single neuron with sigmoid activation produces a probability score in [0, 1]. Compiled with **Adam optimizer**, **binary crossentropy** loss, monitoring accuracy, precision, and recall.

Training Process & Callbacks

Smart Callbacks Ensemble

EarlyStopping

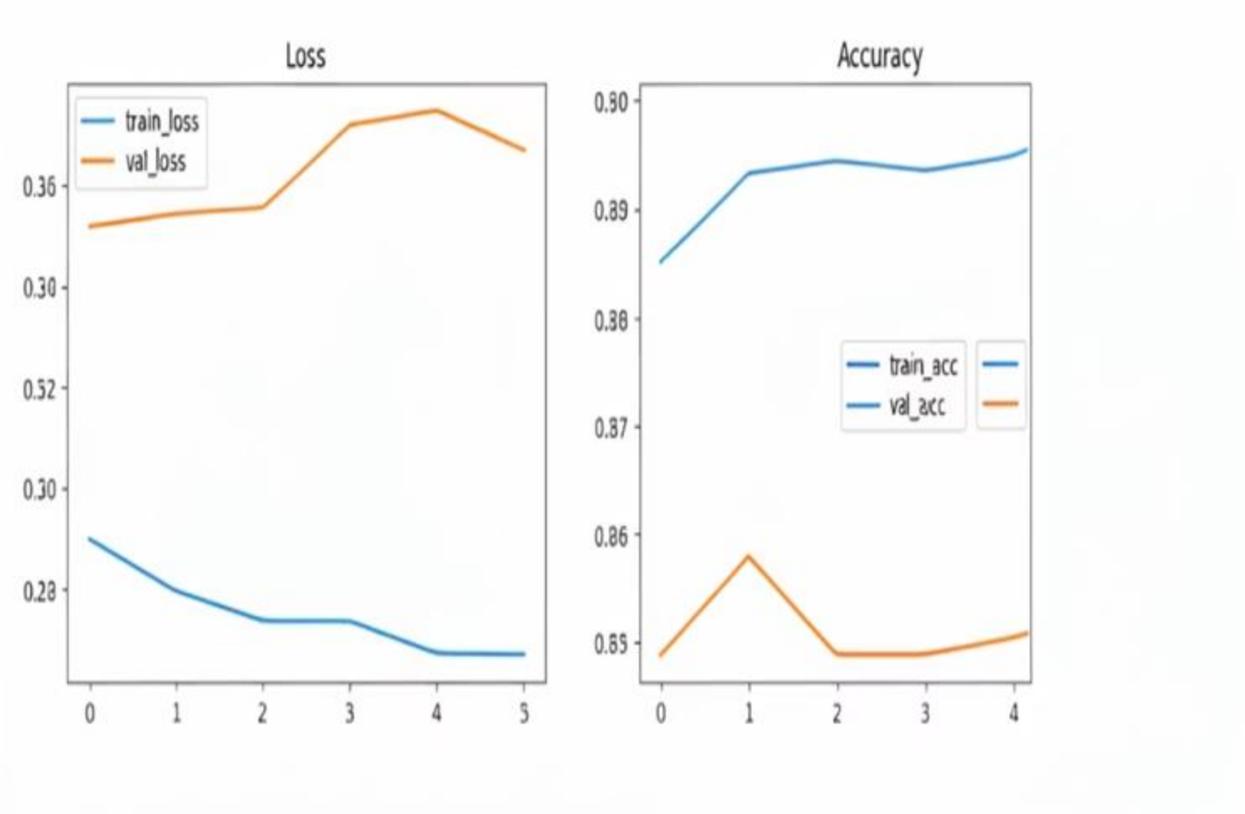
Patience of 5 epochs — training halted automatically at epoch 6, preventing wasted compute and overfitting.

ReduceLROnPlateau

Reduces learning rate by factor 0.2 when validation loss plateaus, enabling smoother convergence.

ModelCheckpoint

Saves the best-performing model weights to disk, ensuring reproducibility and deployment readiness.



Evaluation on the Test Set

88.1%

Test Accuracy

Overall correct classifications on the held-out 1,000-image test set.

96.2%

Recall (Parasitized)

Critically high — the model catches **96 out of 100** infected cells, minimizing dangerous false negatives.

82.8%

Precision

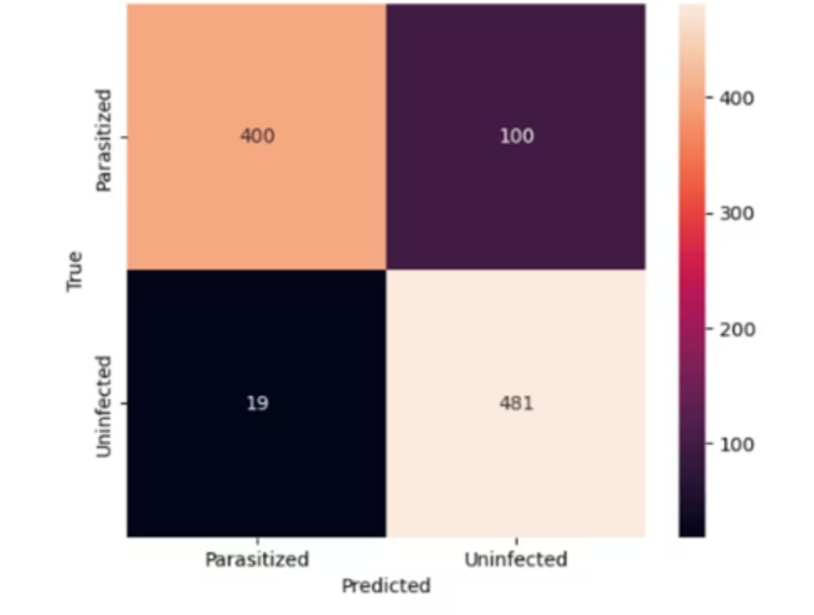
Of all cells predicted as parasitized, 83% are truly infected — a strong positive predictive value.

0.9675

ROC-AUC

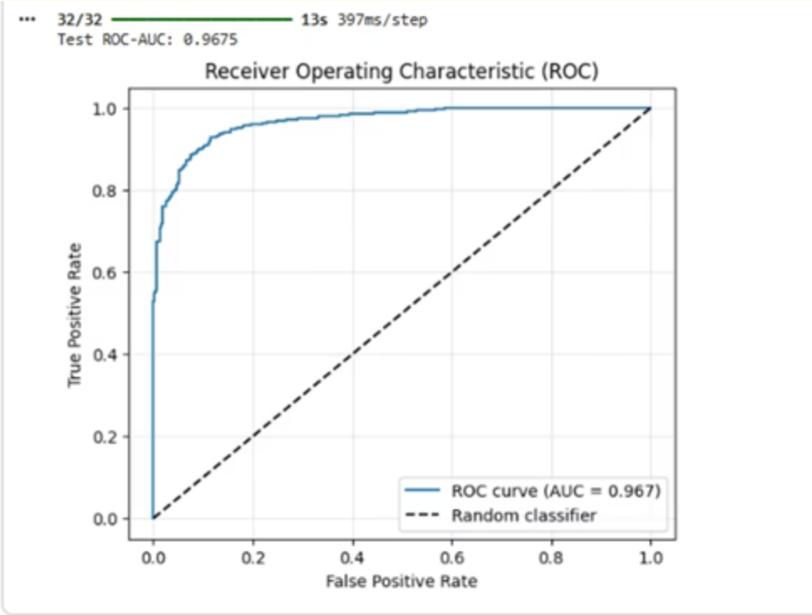
Near-perfect discrimination between classes. An AUC of 0.97 indicates excellent model confidence calibration.

Confusion Matrix



F1-Scores

Parasitized class: **F1 = 0.87**Uninfected class: **F1 = 0.89**



Model Interpretability: Grad-CAM

What Is Grad-CAM?

Gradient-weighted Class Activation Mapping computes the gradient of the class score with respect to the last convolutional layer (`Conv_1` in MobileNetV2). The resulting heatmap highlights the spatial regions that most influenced the prediction.

Clinical Significance

For parasitized cells, the red/yellow activation regions correspond directly to the **Plasmodium parasite's location** inside the erythrocyte — confirming the model is learning biologically meaningful features, not spurious background artifacts.

Why It Matters

Trust: Clinicians can verify the model is focusing on the right cellular structures

Transparency: Reveals failure modes and potential biases in predictions

Adoption: Explainability is a prerequisite for regulatory and clinical acceptance of AI-assisted diagnostics

Conclusion & Future Directions

What We Built

- A **lightweight, accurate malaria detector** using MobileNetV2 transfer learning on NIH cell image data
- **88% test accuracy** and a clinically critical **96% recall** for infected cells, with ROC-AUC of 0.97
- Fully **interpretable predictions** via Grad-CAM heatmaps that align with known parasite morphology

Future Directions

Multi-Class Extension

Expand to distinguish between different Plasmodium species (falciparum, vivax, malariae)

References

Rajaraman, S., et al. (2018). *A comparative study of pre-trained convolutional neural networks for malaria parasite detection*.

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